Session 6: Data visualisation and ggplot

Heiner Salomon

Bremen University

2026-01-06

Universität Bremen

Overview I

1. Introduction
2. Data visualisation
3. ggplot()
4. Any questions?
5. Working on projects

Universität Bremen

Introduction

Learning goals of today

- Short repetition from last week

Universität Bremen

Learning goals of today

- Short repetition from last week
- Crash course in data visualisation

Learning goals of today

- Short repetition from last week
- Crash course in data visualisation
- Intro to ggplot

Repetition: Useful commands

- group_by(): Grouping by variable (for manipulation)

Repetition: Useful commands

- group_by(): Grouping by variable (for manipulation)
- join(): Join two data sets on identifier

Universität Bremen

Repetition: Useful commands

- group_by(): Grouping by variable (for manipulation)
- join(): Join two data sets on identifier
- arrange(): Sort all data by variable(s)

Repetition: Useful commands

- group_by(): Grouping by variable (for manipulation)
- join(): Join two data sets on identifier
- arrange(): Sort all data by variable(s)
- pivot(): Re-order data beween rows and columns

Data visualisation

Why is data visualisation so important?

- There is an abundance of data

Why is data visualisation so important?

- There is an abundance of data
- But all the data in the world is meaningless if it is not understood

Universität Bremen

Why is data visualisation so important?

- There is an abundance of data
- But all the data in the world is meaningless if it is not understood
- Humans are visual creatures

Universität Bremen

Why is data visualisation so important?

- There is an abundance of data
- But all the data in the world is meaningless if it is not understood
- Humans are visual creatures
- Data visualisation is the fastest way for humans to digest complex data

Universität Bremen

Famous people in data viz: Florence Nightingale

- Florence Nightingale

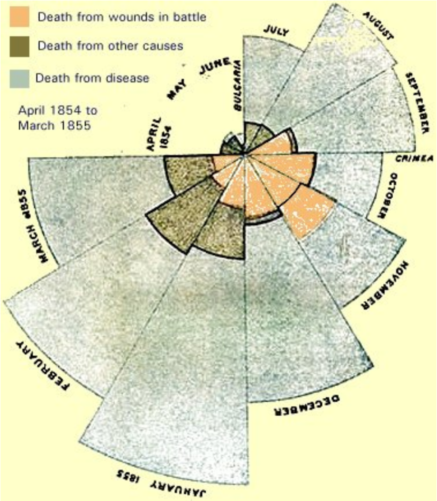
Famous people in data viz: Florence Nightingale

- Florence Nightingale
- Manager and nurse educator in the British Army

Famous people in data viz: Florence Nightingale

- Florence Nightingale
- Manager and nurse educator in the British Army
- Became famous for data visualisation of Crimean War

British Casualties in 1854



Famous people in data viz: Edward Tufte

- American Statistician

Famous people in data viz: Edward Tufte

- American Statistician
- The Visual Display of Quantitative Information

Universität Bremen

Famous people in data viz: Edward Tufte

- American Statistician
- The Visual Display of Quantitative Information
- Famous for data visualisation principles

Some of Tufte's principles for good data vis

- Show the data

Some of Tufte's principles for good data vis

- Show the data
- Maximize data-ink ratio

Universität Bremen

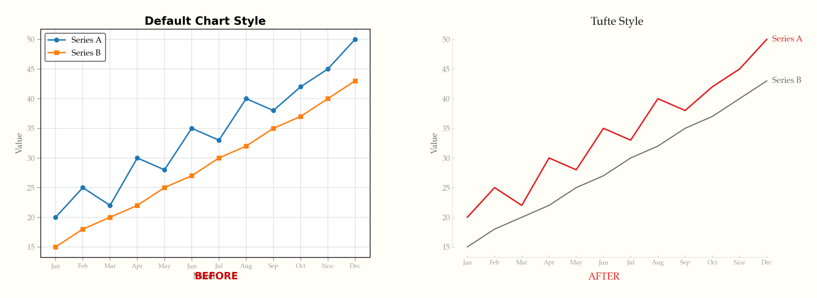
Some of Tufte's principles for good data vis

- Show the data
- Maximize data-ink ratio
- Minimizing chartjunk

Some of Tufte's principles for good data vis

- Show the data
- Maximize data-ink ratio
- Minimizing chartjunk
- Graphical integrity

An example of Tufte's principles



Source: https://raw.githubusercontent.com/caylent/tufte-data-viz/main/_docs/before-after.png

Some further reading

- Data visualization in R

Some further reading

- Data visualization in R
- R Graph Gallery

Some further reading

- Data visualization in R
- R Graph Gallery
- Modern Data Visualization with R

Universität Bremen

Some further reading

- Data visualization in R
- R Graph Gallery
- Modern Data Visualization with R
- Data Visualization in R (Geeks for Geeks)

Universität Bremen

ggplot()

Universität Bremen

Introduction to ggplot()

- ggplot() is the tidyverse function for creating graphs

Introduction to ggplot()

- ggplot() is the tidyverse function for creating graphs
- It is very versatile and provides many options

Universität Bremen

Introduction to ggplot()

- ggplot() is the tidyverse function for creating graphs
- It is very versatile and provides many options
- load using library("ggplot2")

Universität Bremen

Basic components of ggplot() function

- call function: ggplot(...

Basic components of ggplot() function

- call function: ggplot(...
- add data: data = ...

Basic components of ggplot() function

- call function: ggplot(...
- add data: data = ...
- add aesthetics aes(x = ... , y = ... , color = ...)

Universität Bremen

Basic components of ggplot() function

- call function: ggplot(...
- add data: data = ...
- add aesthetics aes(x = ... , y = ... , color = ...)
- add layer that you want to visualise, like points + geom_point()

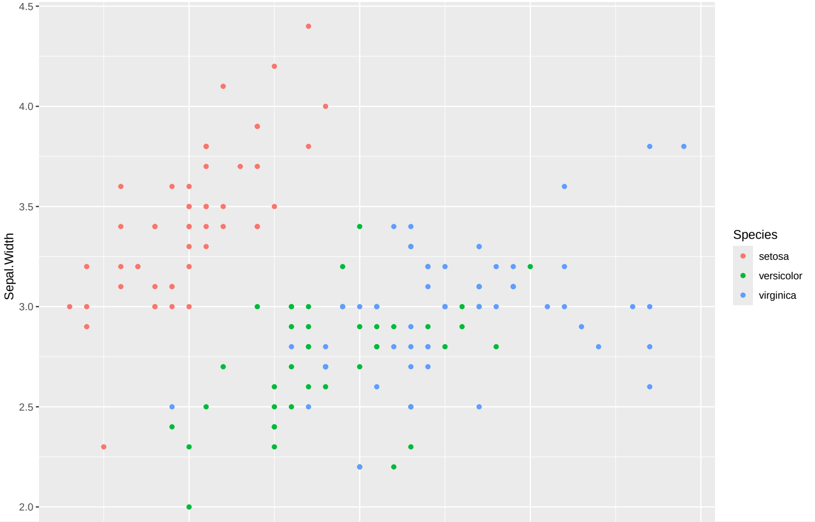
Universität Bremen

Simple example of ggplot() - Code

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set
ggplot(data = iris,
       aes(x = Sepal.Length,
           y = Sepal.Width,
           colour = Species)) +
  geom_point()
```

Universität Bremen

Simple example of ggplot() - Graph



ersität
nen

Task: Simple scatter plot with your data

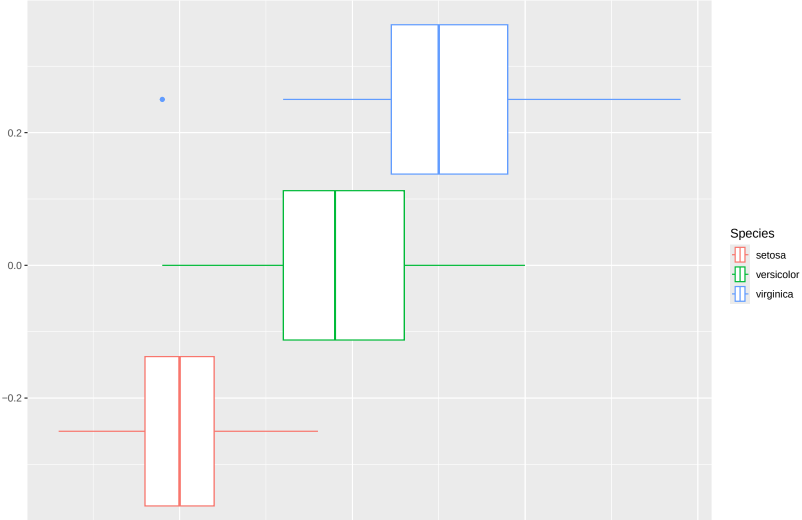
- Try to create two simple scatter plot in your data using ggplot()

```
# Load ggplot2
library(ggplot2)

# Scatterplot using iris data set
ggplot(data = iris,
       aes(x = Sepal.Length,
           y = Sepal.Width,
           colour = Species)) +
  geom_point()
```

Simple example of ggplot() - Alternative code

```
# Load ggplot2
library(ggplot2)

# Boxplot using iris data set
ggplot(data = iris,
       aes(x = Sepal.Length,
           color = Species)) +
  geom_boxplot()
```

Simple example of ggplot() - Alternative graph



There are many types of layers

- I changed the plot entirely by using a different layer

There are many types of layers

- I changed the plot entirely by using a different layer
- Different layers are useful for different

Universität Bremen

List of most important layers

- geom_point() - 2 continuous variables

Universität Bremen

List of most important layers

- geom_point() - 2 continuous variables
- geom_hist() - 1 continuous variables

Universität Bremen

List of most important layers

- geom_point() - 2 continuous variables
- geom_hist() - 1 continuous variables
- geom_boxplot() - 1 continuous variable, one categorical variable

Universität Bremen

List of most important layers

- geom_point() - 2 continuous variables
- geom_hist() - 1 continuous variables
- geom_boxplot() - 1 continuous variable, one categorical variable
- geom_bar() - 1 count variable, one categorical variable

Universität Bremen

Task: Create 4 plots with your data

- geom_point() - 2 continuous variables

Task: Create 4 plots with your data

- geom_point() - 2 continuous variables
- geom_hist() - 1 continuous variables

Universität Bremen

Task: Create 4 plots with your data

- geom_point() - 2 continuous variables
- geom_hist() - 1 continuous variables
- geom_boxplot() - 1 continuous variable, one categorical variable

Task: Create 4 plots with your data

- geom_point() - 2 continuous variables
- geom_hist() - 1 continuous variables
- geom_boxplot() - 1 continuous variable, one categorical variable
- geom_bar() - 1 count variable, one categorical variable

Universität Bremen

Any questions?

Working on projects